Analysis of the Obstacle Potential Function

Cuurent Obtsacle Potential Function

Obstacle potential

$$U_{obst} = \frac{\lambda\left(|\Delta v|/\Delta v_{max}\right)}{k(\Delta x, \Delta y, \Delta v_x, \Delta v_y)}$$

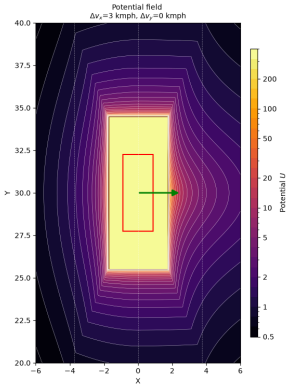
$$k = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \dfrac{w}{|\Delta x|}\right)\sqrt{\left(\dfrac{\Delta x}{\tau(\Delta x, \Delta v_x)}\right)^2 + \left(\dfrac{\Delta y}{\tau(\Delta y, \Delta v_y)}\right)^2}, & |\Delta y| \leq \dfrac{l\,|\Delta x|}{w} \\ \left(1 - \dfrac{l}{|\Delta y|}\right)\sqrt{\left(\dfrac{\Delta x}{\tau(\Delta x, \Delta v_x)}\right)^2 + \left(\dfrac{\Delta y}{\tau(\Delta y, \Delta v_y)}\right)^2}, & \text{otherwise} \end{cases}$$

$$\tau(\Delta x, \Delta v_x) = \tfrac{1}{2}\Big[(1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1)\tanh(-\Delta x \Delta v_x)\Big]\,\tau_x$$

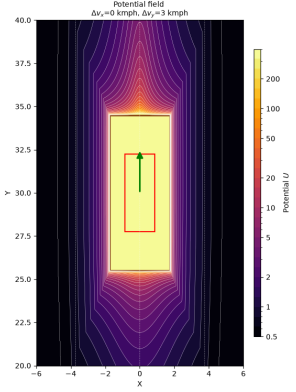
$$\tau(\Delta y, \Delta v_y) = \tfrac{1}{2}\Big[(1 + \alpha|\Delta v_y|) + (\alpha|\Delta v_y| - 1)\tanh(-\Delta y \Delta v_y)\Big]\,\tau_y$$

$\Delta x, \Delta y$: position relative to the obstacle vehicle; $\Delta v_x, \Delta v_y$: velocity relative to the obstacle vehicle.

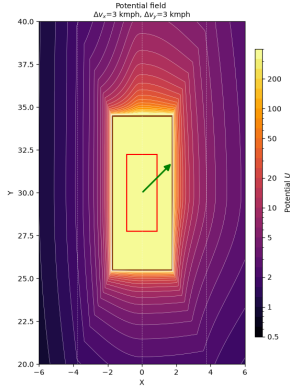
Potential field U_{obst}



Varying Δv_x

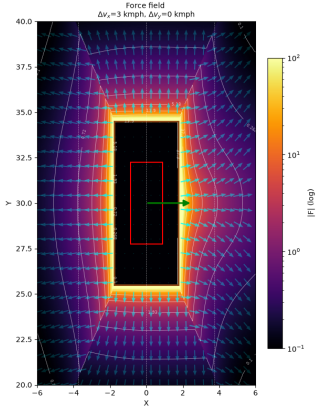


Varying Δv_y

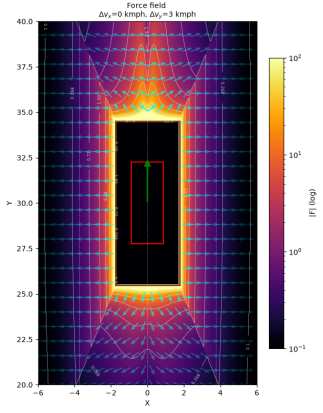


Varying $\Delta v_x, \Delta v_y$

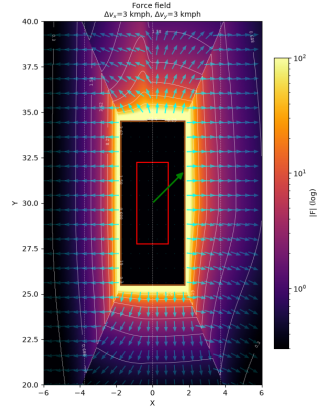
Force field $F = -\nabla U_{obst}$



Varying Δv_x



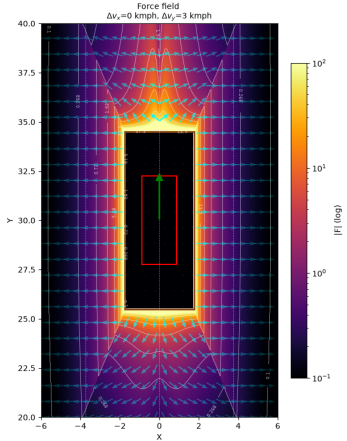
Varying Δv_y



Varying $\Delta v_x, \Delta v_y$

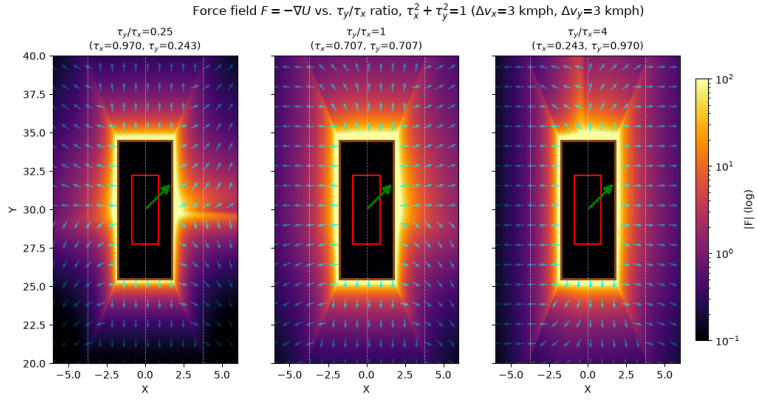
Possible Issues

Issue 1: Anomalous dip in force magnitude



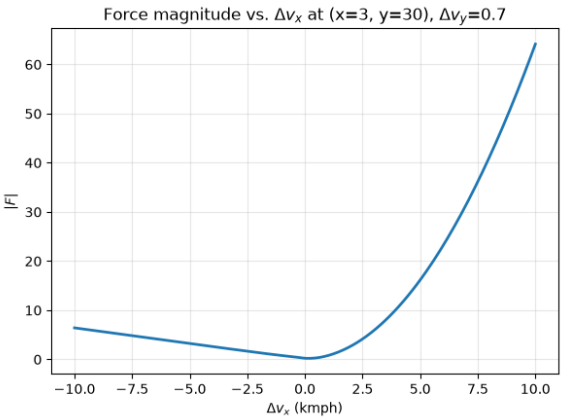
- The force magnitude **dips near** $x = 0$.
- Undesirable: along $x = 0$ the force should be *maximal* compared with neighbouring points.

Issue 2: Sensitivity to τ_x, τ_y



- The shape of the potential is highly sensitive to τ_y/τ_x.
- Very large or very small ratios make the force field **parallel** in some regions.

Issue 3: Velocity behaviour at a fixed point



- Force **increases** with receding velocity at a fixed point.
- Ideally the repulsive force should **decrease** when the vehicles are receding.

Issue 4: More free parameters than DOF

- The model has **4 free parameters**: λ, α, τ_x, τ_y.
- The model only has **3 degrees of freedom**.
- Reducing the parameter count will help the optimisation problem.

Potential Fixes

Force dip: the cause

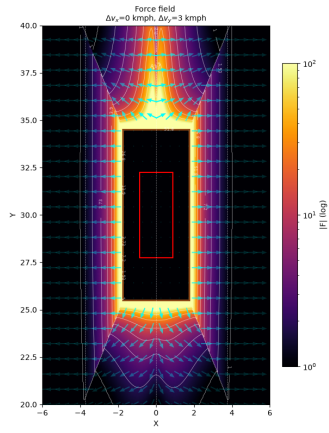
The dip comes from the skewed distance

$$k = \sqrt{(x/\tau_x)^2 + (y/\tau_y)^2}$$

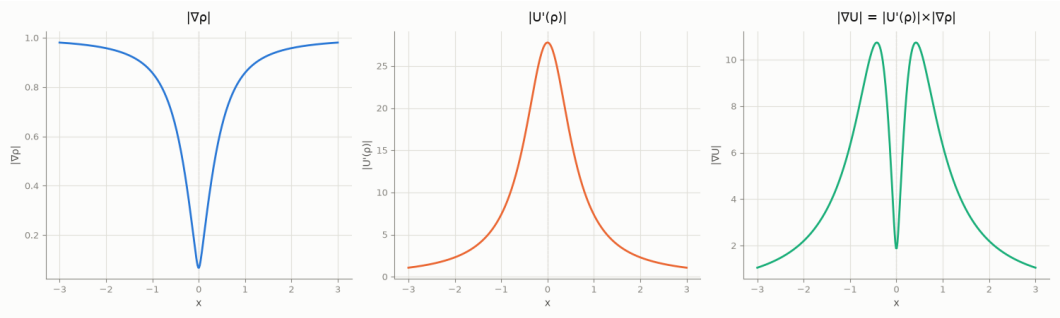
$$U(k) = \frac{\lambda\left(|\Delta v|/\Delta v_{max}\right)}{k}, \qquad F = -\nabla U$$

$$\nabla U = U'(k)\,\nabla k \quad \Rightarrow \quad |\nabla U| = |U'(k)| \cdot |\nabla k|$$

$|\nabla k|$ is not uniform, which creates the dip.



Force dip: the cause



$$|\nabla U| = |U'(k)| \cdot |\nabla k|$$

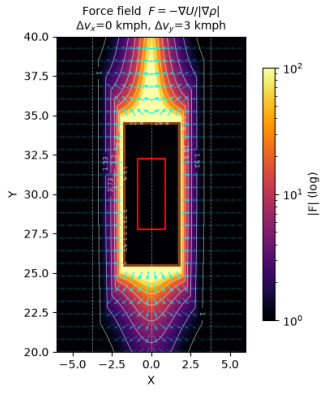
Force dip: the fix

$$U'(k) = -\frac{\lambda\left(|\Delta v|/\Delta v_{max}\right)}{k^2}$$

$U'(k)$ falls off smoothly with k and has no dip or ridge of its own.

Normalise the gradient direction:

$$F = -\frac{\nabla U}{|\nabla k|} = -U'(k)\,\frac{\nabla k}{|\nabla k|}, \qquad |F| = |U'(k)|$$



τ sensitivity: the cause

$$U = \frac{\lambda(\Delta v/v)}{k}$$

$$k = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \dfrac{w}{|\Delta x|}\right)\sqrt{\left(\dfrac{\Delta x}{\tau(\Delta x, \Delta v_x)}\right)^2 + \left(\dfrac{\Delta y}{\tau(\Delta y, \Delta v_y)}\right)^2}, & |\Delta y| \leq \dfrac{l|\Delta x|}{w} \\ \left(1 - \dfrac{l}{|\Delta y|}\right)\sqrt{\left(\dfrac{\Delta x}{\tau_x}\right)^2 + \left(\dfrac{\Delta y}{\tau_y}\right)^2}, & \text{otherwise} \end{cases}$$

- If $\tau_y \ll \tau_x$, the y^2/τ_y^2 term dominates.
- The potential then does not change across $x \Rightarrow$ **no lateral force** in the forward and backward regions.
- The same happens in the side region when $\tau_y/\tau_x \gg 1$.

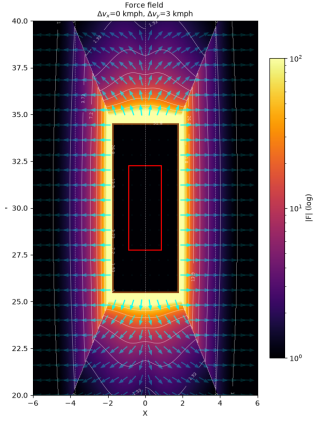
τ sensitivity: the fix

k already splits space into regions, so use **different τ per region**:

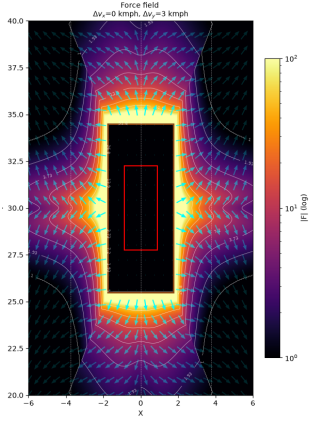
$$k = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \dfrac{w}{|\Delta x|}\right) \sqrt{\left(\dfrac{\Delta x}{\tau(\Delta x, \Delta v_x, \tau_2)}\right)^2 + \left(\dfrac{\Delta y}{\tau(\Delta y, \Delta v_y, \tau_1)}\right)^2}, & |\Delta y| \leq \dfrac{l|\Delta x|}{w} \\ \left(1 - \dfrac{l}{|\Delta y|}\right) \sqrt{\left(\dfrac{\Delta x}{\tau(\Delta x, \Delta v_x, \tau_1)}\right)^2 + \left(\dfrac{\Delta y}{\tau(\Delta y, \Delta v_y, \tau_2)}\right)^2}, & \text{otherwise} \end{cases}$$

$$\tau(\Delta x, \Delta v_x, \tau) = \frac{\tau}{2}\Big[(2 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x|)\tanh(-\Delta x \Delta v_x)\Big]$$

τ sensitivity: comparison

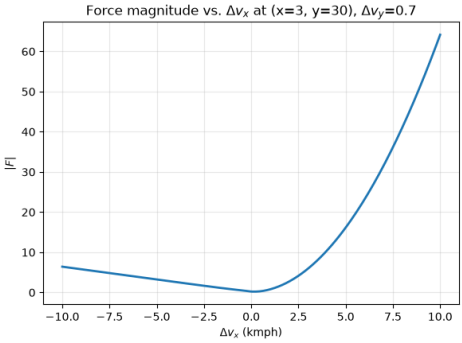


Before: $\tau_y/\tau_x > 5$

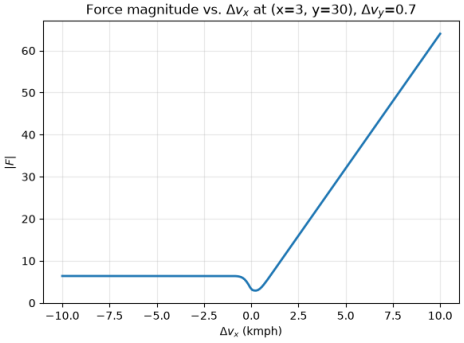


After: $\tau_2/\tau_1 > 5$

Velocity behaviour: the issue



Original



After removing $|\Delta v|/\Delta v_{max}$

- Force grows with receding velocity because of the $|\Delta v|/\Delta v_{max}$ term in the numerator.
- Fix is to remove that term, $U(k) = \lambda/k$.
- The remaining kink comes from the τ term.

Reformulating τ

Current form:

$$\tau = \Big[(1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1)\tanh(-\Delta x \Delta v_x)\Big]\,\tau_x$$

Original goal:

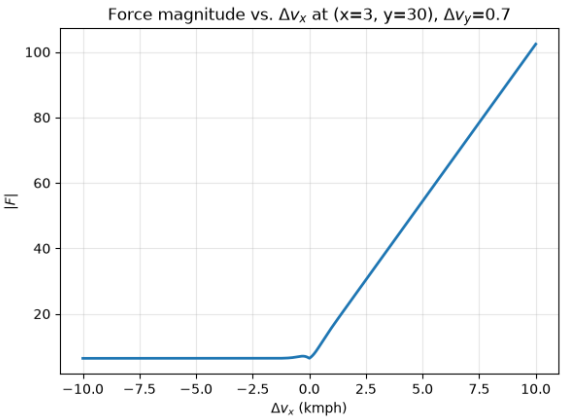
$$\tau = \tau_x \text{ (receding)}, \qquad \tau = \alpha\tau_x|\Delta v_x| \text{ (approaching)}$$

Better: τ should be larger when vehicles approach:

$$\tau = \tau_x \text{ (receding)}, \qquad \tau = \tau_x(1 + \alpha|\Delta v_x|) \text{ (approaching)}$$

Formulation 1

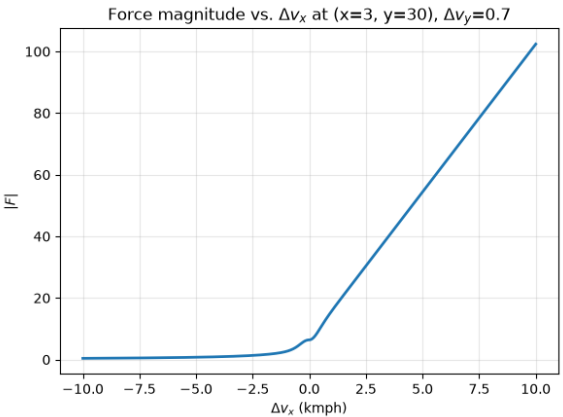
$$\tau = \Big[\big(1 + (1 + \alpha|\Delta v_x|)\big) + \big((1 + \alpha|\Delta v_x|) - 1\big)\tanh(-\Delta x \Delta v_x)\Big]\,\tau_x$$



- Force magnitude vs. Δv_x.
- We still want the repulsive force to **decrease** with increasing receding velocity.

Formulation 2

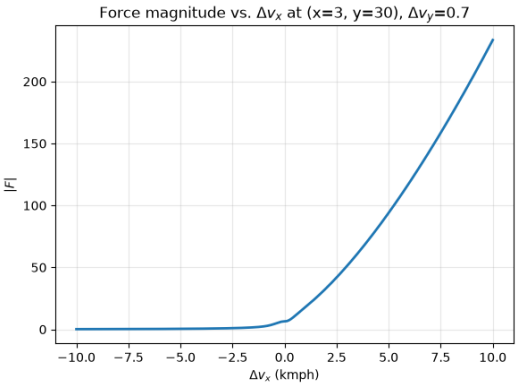
$$\tau = \left[\left(\frac{1}{1+\alpha|\Delta v_x|} + (1+\alpha|\Delta v_x|)\right) + \left((1+\alpha|\Delta v_x|) - \frac{1}{1+\alpha|\Delta v_x|}\right)\tanh(-\Delta x \Delta v_x)\right]\tau_x$$



- For high approaching velocity, $\tau \approx \alpha|\Delta v_x|\,\tau_x$.
- α and τ_x become redundant.
- Use α instead to control the **power** of the velocity term.

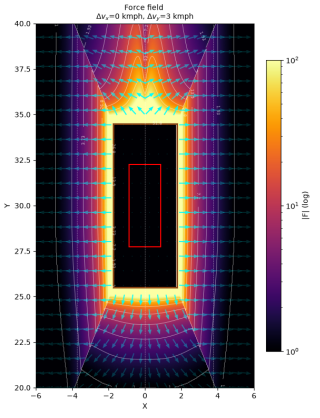
Formulation 3

$$\tau = \left[\left(\frac{1}{(1+|\Delta v_x|)^\alpha} + (1+|\Delta v_x|)^\alpha\right) + \left((1+|\Delta v_x|^\alpha) - \frac{1}{(1+|\Delta v_x|)^\alpha}\right)\tanh(-\Delta x \Delta v_x)\right]\tau_x$$

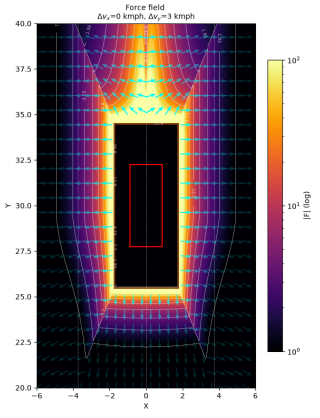


Force magnitude vs. Δv_x under Formulation 3

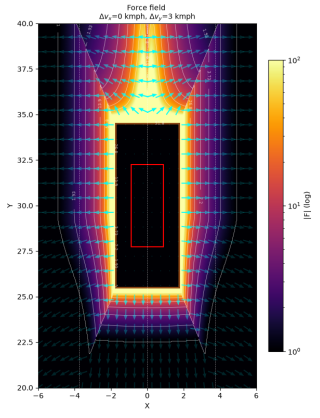
Formulations compared



Formulation 1



Formulation 2



Formulation 3

Fix 4 — Reducing free parameters

Define

$$\gamma = \frac{\tau_y}{\tau_x}, \qquad \beta = \lambda \tau_x$$

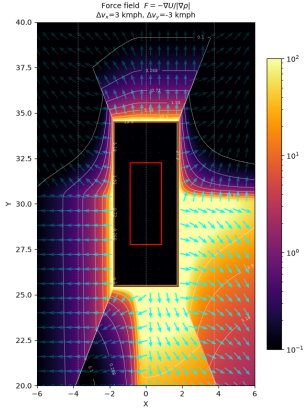
The equation reduces to

$$U = \frac{\beta(\Delta v/v)}{k}, \qquad k = \begin{cases} 0, & |\Delta y| < l \text{ and } |\Delta x| < w \\ \left(1 - \dfrac{w}{|\Delta x|}\right) \sqrt{(\Delta x)^2 + \left(\dfrac{\Delta y}{\tau(\gamma)}\right)^2}, & |\Delta y| \leq \dfrac{l|\Delta x|}{w} \\ \left(1 - \dfrac{l}{|\Delta y|}\right) \sqrt{(\Delta x)^2 + \left(\dfrac{\Delta y}{\tau(\gamma)}\right)^2}, & \text{otherwise} \end{cases}$$

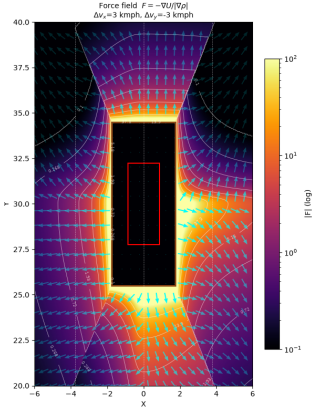
- Free parameters reduced from λ, τ_x, τ_y to γ, β.

Combining the changes

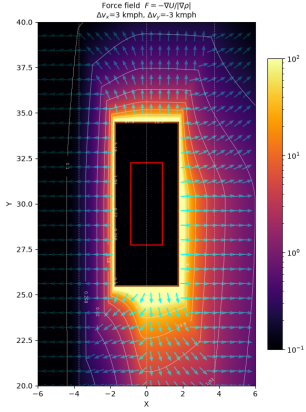
Final combinations



All modifications combined



Formulation 1



Formulation 1 without τ correction